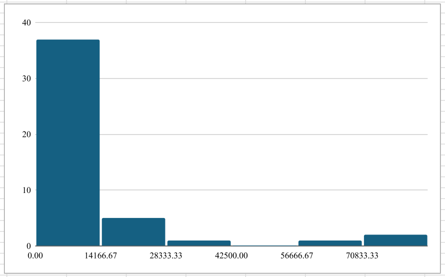
State	Region	U.S. certified organic livestock, 2008, by State						U.S. certified organic livestock, 2011, by State						Random Va
		Cows, pigs, and sheep						Cows, pigs, and sheep						
		Beef cows	Milk cows	Other cows	Hogs & pigs	Sheep & lambs	Total	Beef cows	Milk cows	Other cows	Hogs & pigs	Sheep & lambs	Total	
		Number						Number						
U.S.		63,680	249,766	144,817	10,111	7,455	475,829	106,181	254,771	113,114	12,373	5,914	492,353	
South Dakota	Midwest	3,964	199	1,398	46	146	5,753	3,991	364	1,416	24		5,795	0
Nevada	West	-	-	-	-	-	-	797					797	0
Tennessee	South	-	-	25	-	-	25		194	337	57		588	0
Wyoming	West	3,996	-	5,041	-	-	9,037	5,216	398	3,832			9,446	0
Idaho	West	2,751	5,280	6,425	11	-	14,467	3,793	3,628	685	181	1,281	9,568	0
Louisiana	South	-	540	387	-	24	951	2	960	332		47	1,341	0
Iowa	Midwest	3,602	4,702	3,663	3,961	1,491	17,419	1,897	5,106	2,344	4,406	490	14,243	0
Texas	South	5,267	26,727	7,541	-	-	39,535	45,790	24,101	5,396	18	837	76,142	0
Ohio	Midwest	533	7,853	4,999	205	122	13,712	874	7,225	3,587	12	3	11,701	0
New Hampshire	Northeast	51	502	349	-	-	902	52	990				1,042	0
New Jersey	Northeast	73	43	33	46	-	195	305	3		1,000		1,308	0
Florida	South	-	-	-	-	-	-			94			94	0
Nebraska	Midwest	6,213	903	4,868	13	80	12,077	3,287	191	595	20		4,093	0
Washington	West	2,000	9,022	7,022	360	-	18,404	17	6,325	4,592			10,934	0
Arkansas	South	23	747	273	-	-	1,043							0
Montana	West	1,446	-	430	682	686	3,244	1,645	229		555	580	3,009	0
California	West	13,177	55,224	8,265	1,803	302	78,771	14,024	57,809	7,594	1,810	594	81,831	0
Michigan	Midwest	722	2,306	2,565	23	268	5,884	1,170	3,722	1,515	28	337	6,772	0
Rhode Island	Northeast	14	16	25	11	-	66	41	16	25	11	15	108	0
North Carolina	South	10	145	117	16	30	318	938	719	504			2,161	0
Delaware	South	19	-	28	-	-	47	12		12			24	0
Vermont	Northeast	1,131	12,260	9,527	43	89	23,050	1,020	12,868	10,831	64	349	25,132	0
Wisconsin	Midwest	3,195	25,764	21,311	1,139	118	51,527	2,216	31,874	25,651	1,933	561	62,235	0
North Dakota	Midwest	1,700	33	1,908	22	-	3,663	1,880	5	602			2,487	0
Utah	West	269	80	-	-	-	349	135	226	98	4		463	0
Oregon	West	1,170	16,728	17,384	20	508	35,810	2,316	11,833	10,691	22	420	25,282	0
Kentucky	South	150	1,600	-	-	-	1,750		1,175	39	25		1,239	0
Oklahoma	South	572	-	1,098	-	-	1,670	221	224	350			795	0
Arizona	West	-	760	248	-	-	1,008	187	661	101			949	0
Illinois	Midwest	425	944	1,047	91	30	2,537	825	1,146	684	136	15	2,806	0
Hawaii	West	205	-	-	-	-	205	25					25	0
Connecticut	Northeast	-	-	-	-	-	-							0
Maine	Northeast	2,192	4,047	4,454	232	251	11,176	635	3,650	2,582	262	144	7,273	1
Missouri	Midwest	1,128	1,106	507	178	111	3,030	727	1,050	220	40		2,037	1
Colorado	West	781	5,370	331	-	2,229	8,711	2,775	10,369	179	45		13,368	1
Alabama	South	-	-	-	-	-	-	17	440				457	1
South Carolina	South	-	-	-	-	-	-	125					125	1
Indiana	Midwest	246	4,703	2,894	196	85	8,124	164	6,503	1,016	368	20	8,071	1
Pennsylvania	Northeast	605	11,614	7,857	72	60	20,208	912	15,992	8,109	35	95	25,143	1
Maryland	South	79	1,153	707	-	127	2,066	158	1,070	919		65	2,212	1
Minnesota	Midwest	1,759	8,142	6,350	140	139	16,530	2,285	11,672	9,809	194	54	24,014	1
New Mexico	West	550	5,350	8,103	-	300	14,303	3,075	1,104	1,095			5,274	1
Virginia	South	165	865	812	19	-	1,861	167	1,280	780	316		2,543	1
West Virginia	South	-	-	-	-	185	185		79	77			156	1
Alaska	West	-	-	-	-	-	-							1
Massachusetts	Northeast	45	354	375	-	2	776	32	694	116	60	3	905	1
Georgia	South	-	-	-	-	-	-							1
New York	Northeast	2,267	34,443	6,299	206	72	43,287	1,879	28,446	6,035	339	4	36,703	1
Kansas	Midwest	994	-	151	576	-	1,721	523			408		931	1
Mississippi	South	191	241	-	-	-	432	31	430	270			731	1

Label / Event	Google	Value /
Prob of State in West	**COUNTIF(B9:**	26% (0.26)
Prob of 2011 Total	COUNTIF(N9:	24% (0.24)
Prob of 2011 Milk	COUNTIFS(J9:	26% (0.26)



Sample 10	State	Region	U.S. certified organic livestock, 2008, by State						U.S. certified organic livestock, 2011, by State						Random Valu
			Cows, pigs, and sheep						Cows, pigs, and sheep						
			Beef cows	Milk cows	Other cows	Hogs & pigs	Sheep & lambs	Total	Beef cows	Milk cows	Other cows	Hogs & pigs	Sheep & lambs	Total	
			Number						Number						
	U.S.		70,320	198,149	116,028	12,604	5,139	402,240	106,181	254,771	113,114	12,373	5,914	492,353	
	South Dakota	Midwest	3,964	199	1,398	46	146	5,753	3,991	364	1,416	24		5,795	0
	Nevada	West	-	-	-	-	-	-	797					797	0
	Tennessee	South	-	-	25	-	-	25		194	337	57		588	0
	Wyoming	West	3,996	-	5,041	-	-	9,037	5,216	398	3,832			9,446	0
	Idaho	West	2,751	5,280	6,425	11	-	14,467	3,793	3,628	685	181	1,281	9,568	0
	Louisiana	South	-	540	387	-	24	951	2	960	332		47	1,341	0
	Iowa	Midwest	3,602	4,702	3,663	3,961	1,491	17,419	1,897	5,106	2,344	4,406	490	14,243	0
	Texas	South	5,267	26,727	7,541	-	-	39,535	45,790	24,101	5,396	18	837	76,142	0
	Ohio	Midwest	533	7,853	4,999	205	122	13,712	874	7,225	3,587	12	3	11,701	0
	New Hampshire	Northeast	51	502	349	-	-	902	52	990				1,042	0
Sample 25	South Dakota	Midwest	3,964	199	1,398	46	146	5,753	3,991	364	1,416	24		5,795	0
	Nevada	West	-	-	-	-	-	-	797					797	0
	Tennessee	South	-	-	25	-	-	25		194	337	57		588	0
	Wyoming	West	3,996	-	5,041	-	-	9,037	5,216	398	3,832			9,446	0
	Idaho	West	2,751	5,280	6,425	11	-	14,467	3,793	3,628	685	181	1,281	9,568	0
	Louisiana	South	-	540	387	-	24	951	2	960	332		47	1,341	0
	Iowa	Midwest	3,602	4,702	3,663	3,961	1,491	17,419	1,897	5,106	2,344	4,406	490	14,243	0
	Texas	South	5,267	26,727	7,541	-	-	39,535	45,790	24,101	5,396	18	837	76,142	0
	Ohio	Midwest	533	7,853	4,999	205	122	13,712	874	7,225	3,587	12	3	11,701	0
	New Hampshire	Northeast	51	502	349	-	-	902	52	990				1,042	0
	New Jersey	Northeast	73	43	33	46	-	195	305	3		1,000		1,308	0
	Florida	South	-	-	-	-	-	-			94			94	0
	Nebraska	Midwest	6,213	903	4,868	13	80	12,077	3,287	191	595	20		4,093	0
	Washington	West	2,000	9,022	7,022	360	-	18,404	17	6,325	4,592			10,934	0
	Arkansas	South	23	747	273	-	-	1,043							0
	Montana	West	1,446	-	430	682	686	3,244	1,645	229		555	580	3,009	0
	California	West	13,177	55,224	8,265	1,803	302	78,771	14,024	57,809	7,594	1,810	594	81,831	0
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	Rhode Island	Northeast	14	16	25	11	-	66	41	16	25	11	15	108	0
	North Carolina	South	10	145	117	16	30	318	938	719	504			2,161	0
	Delaware	South	19	-	28	-	-	47	12		12			24	0
	Vermont	Northeast	1,131	12,260	9,527	43	89	23,050	1,020	12,868	10,831	64	349	25,132	0
	Wisconsin	Midwest	3,195	25,764	21,311	1,139	118	51,527	2,216	31,874	25,651	1,933	561	62,235	0
	North Dakota	Midwest	1,700	33	1,908	22	-	3,663	1,880	5	602			2,487	0
	Utah	West	269	80	-	-	-	349	135	226	98	4		463	0

50	
Total states = 50	
States in the west region	13
div west states/total states	0.26
What is the probability tha	0.26
What is the probability	0.2448979592
stateswith livestock>10k	12
total states	50
"What is the probability tha	0.24

What is the probability that a randomly selected state has between 1,000 and 10,000 Milk Cows in 2011?

milk cows btwn 1k & 10k	13
Total states	50
Probability	0.26
Alternatively	0.2653061224

Plain Language Explanation:
"Nearly a quarter of all states (24%) operate on a large scale with more than
10,000 total certified organic livestock head in 2011.
That means about 12 out of 50 states have major organic livestock operations e

3. 2011 Milk Cows Between 1,000 and 10,000 (26\%)
Plain Language Explanation:
"About 1 out of every 4 states (26%) maintains a mid-sized organic dairy prese
meaning they have at least 1,000 milk cows but no more
than 10,000. Exactly 13 states fall into this mid-tier milk production range."

1. Population Mean Formu	10,703
Sample MEAN 1 ($n = 10	13,066
Sample MEAN 2 ($n = 25	13,805

Texas (76,142), and Wisconsin (62,235) having massive counts—
if your random draw happened to include two or three of these high-volume states,
it significantly pulls the sample averages upward.

Sampling Variability in Skewed Data:
In right-skewed datasets, smaller random samples often experience higher variability.
Drawing a sample that catches a few huge outliers will yield a much higher mean,
while drawing a sample that misses them will yield a much lower mean.

When writing up your analysis:
State that both samples pulled higher averages due to sampling variability and
the inclusion of extreme outlier states in the random draws.

Highlight that in highly skewed agricultural data, sampling error is common,
which is why larger sample sizes or stratified sampling are often used in practice.

Statistic	sample(n = 10)	sample(n = 25)
Sample Mean	13,066	13,805
Sample Std Dev	22726.19443	23944.7188
Standard Error (SE)	7186.653695	4788.94377
Margin of Error (ME)	14085.84124	9386.32979
Lower Bound (LCB)	(1,020)	4,418
Upper Bound (UCB)	27,152	23,191

What Changed as the Sample Size Increased?
Increasing your sample size from 10 to 25 reduced the uncertainty by shrinking your interval width from 28,172 down to 18,772, making your estimate significantly more precise!

1. The Standard Error Shrinks
Because sample size (n) sits in the denominator of the standard error formula (
n

s

), taking the square root of a larger number (
25

=5 vs
10

≈3.16) makes the resulting Standard Error smaller.

2. The Margin of Error Shrinks
Since Margin of Error=1.96×SE, a smaller Standard Error directly leads to a smaller Margin of Error.

3. The Confidence Interval Becomes Narrower (More Precise)
Narrower Interval: A smaller margin of error pulls the lower and upper bounds closer together.

Higher Precision: A narrower interval means your estimate of the population mean is more precise and tightly bounded.

1. The Mathematical Explanation
The overall width of a confidence interval is determined entirely by the Margin of Error (ME):

Margin of Error=Z×Standard Error=Z×
n

s

Where:

Z is the confidence factor (1.96 for 95% confidence).

s is the sample standard deviation.

n is the sample size.

The Power of the Denominator (
n

)
Notice where sample size (n) sits in the formula: it is in the denominator under a square root (
n

).

For n=10: You divide the standard deviation s by
10

≈3.16.

For n=25: You divide the standard deviation s by
25

=5.00.

Because you are dividing by a larger number (5.00 vs 3.16), the resulting Standard Error becomes smaller.

Since the Margin of Error is simply 1.96×SE, making the Standard Error smaller directly forces the Margin of Error—and thus the entire confidence interval width—to shrink.

Math Snapshot: Increasing n from 10 to 25 increases the denominator by a factor of
10

25

=
3.16
5

≈1.58, reducing your margin of error by roughly 36.8% (assuming standard deviation stays roughly similar).

2. What a Narrower Interval Means for Reliability
A narrower confidence interval directly impacts precision, uncertainty, and decision-making reliability:

Higher Precision
Wide Interval (n=10): Saying the average state organic livestock count is somewhere between −1,020 and 27,152 is so broad that it gives you very little actionable information.

Narrow Interval (n=25): Pinning the average down to between 4,419 and 23,191 narrows the "range of plausible values" significantly.

Reduced Sampling Uncertainty
Because larger samples capture more information from the population, there is less chance that a few random outliers will severely distort your estimate. The smaller Standard Error reflects that sample means fluctuate much less when sample sizes are larger.

Better Practical Utility
In financial or resource-allocation contexts (such as distributor subsidies for organic farms), a narrower interval gives stakeholders far greater confidence when budgeting, as the margin of error surrounding expected averages is much smaller. Comparative Analysis & Interpretation of Results

1. Population Data vs. Sample Data
Population Data: Refers to the full dataset containing every single U.S. state in 2011. Its mean (μ=10,703) represents the true average livestock count across all states.

Sample Data: Refers to a randomly chosen subset of states (n=10 and n=25). Because we only look at a portion of the total data, sample means will naturally vary from the true population mean due to sampling error.

2. What a 95% Confidence Level Means
A 95% confidence level does not mean there is a 95% chance a specific interval contains the true mean. Instead, it describes the reliability of the process:

If we were to repeat this random sampling process 100 times and calculate a confidence interval each time, roughly 95 out of those 100 intervals would successfully capture the true population mean (μ=10,703).

3. What the Margin of Error Tells Us About Precision
The margin of error defines the buffer of uncertainty around our sample estimate:

A larger margin of error (±14,086 for n=10) indicates lower precision and high uncertainty, creating a very wide interval ([−1,020,27,152]).

A smaller margin of error (±9,386 for n=25) indicates higher precision, giving us a tighter range of plausible values ([4,419,23,191]).

4. Why Larger Samples Produce Narrower Confidence Intervals
Because sample size (n) sits in the denominator of the standard error formula (SE=
n

s

), collecting more data lowers the standard error.

As sample size increases, random variations and extreme outliers balance out, giving us more information about the population.

Smaller standard errors directly shrink the margin of error, resulting in a narrower, more precise confidence interval.

5. The Role of the Z-Score (1.96)

The Z-score acts as a scaling multiplier based on the normal distribution:

It tells us how many standard errors away from the sample mean we must extend to cover 95% of the area under a standard normal curve.

Multiplying the standard error by 1.96 guarantees that our interval expanded far enough to meet our 95% statistical confidence target.

1. What Happens to the Margin of Error and Confidence Interval?

If you switch from a 95% confidence level to a 99% confidence level:

The Margin of Error gets LARGER.

The Confidence Interval gets WIDER.

2. What Z-Score Would You Use?

For a 99% confidence level, you would use a Z-score of approximately 2.576 (often rounded to 2.58 in textbook calculations).

3. Why Does a Higher Confidence Level Produce a Wider Interval?

There is an inherent trade-off in statistics between precision and certainty:

The Net Analogy: Think of a confidence interval like throwing a net to catch a fish (the true population mean). If you want to be 95% sure you catch the fish, you need a net of a certain size. But if you want to be 99% sure you catch it, you have to throw a much wider net.

Mathematical Explanation: In the margin of error formula:

Margin of Error=Z×Standard Error

Because 2.576 is larger than 1.96, multiplying your standard error by 2.576 increases the margin of error. Adding and subtracting this larger margin of error pushes your lower and upper bounds further apart.

Comparison Table (n=25 Example)

Metric	95% Confidence Level	99% Confidence Level
Z-Score	1.96	2.576
Margin of Error Formula	1.96×4,788.94	2.576×4,788.94
Margin of Error	9,386.33	12,336.31
Lower Bound	4,419	1,469
Upper Bound	23,191	26,141
Interval Width	18,772	24,672 (Wider)

Summary Rule: Higher confidence gives you more certainty, but costs you precision by widening the interval.

1. How Using a Sample Changed Confidence & Which Sample Size Felt More Reliable

Population vs. Sample Confidence: Working with the full population gave complete certainty—we knew the exact average was 10,703. Moving to a sample introduced uncertainty (sampling error), meaning we could no longer state a single exact average with 100% certainty. Instead, we had to rely on a range of plausible values (the confidence interval).

Which Sample Size Felt More Reliable: The sample size of 25 (n=25) felt significantly more reliable.

The small sample (n=10) produced an interval so wide (−1,020 to 27,152) that it even dipped into negative numbers, making it practically useless for real-world planning.

The larger sample (n=25) produced a much tighter interval (4,419 to 23,191) centered around a very similar average, giving far more confidence that the true population average was being captured realistically.

2. Relationship Between Sample Size and Margin of Error

Inverse Relationship: As the sample size increased, the margin of error decreased.

The Math Behind It: In Google Sheets, because sample size (n) sits inside the square root in the denominator of the standard error formula (=STDEV / SQRT(n)), a larger sample size forces the standard error down.

Impact on the Calculations:

For n=10, the margin of error was 14,086.

For n=25, the margin of error dropped by over 33% down to 9,386.

Takeaway: More data points dilute the influence of extreme outlier states (like California or Texas), shrinking the buffer of uncertainty around our estimate.

3. Explaining the n=25 Confidence Interval to a Non-Technical Stakeholder

*"Based on our study of 25 randomly selected states, we estimate that a typical U.S. state manages around 13,800 certified organic livestock.

Because we only looked at a sample rather than counting every state in the country, we built in a safety buffer. We are 95% confident that the true national average across all states falls somewhere between 4,400 and 23,200 livestock head.

In simple terms, while state counts vary widely across the nation, this range gives us a realistic and reliable ballpark for planning our food sustainability initiatives without having to audit all 50 states."*